PHYS 135B Discussion 8

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I. The magnetic field of a solenoid (HW6 Q12 (a))

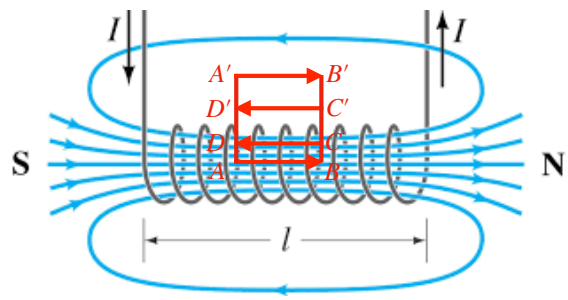

Figure 1: The magnetic field of a long solenoid

First of all, there is no radial component of **B** due to the symmetry[1]. Then for the loop $A'B'C'D'$

$$\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = [B_{A'B'} - B_{D'C'}]\Delta l = 0$$

with the subscrips being the line segments on which the **B** field is defined. Therefore, the field outside does not depend on the distance from the axis. But we know that the field at the infinite should be zero. As a result, it must be zero everywhere outside the solenoid.

For the loop $ABC'D'$ with AB being an a line segment inside the solenoid. Then,

$$\oint \mathbf{B} \cdot \mathrm{d}\mathbf{l} = [B_{AB} - B_{D'C'}]\Delta l = 4\pi n \Delta l I/c \quad \text{(cgs)}$$

$$B_{AB} = \frac{4\pi n I}{c},$$

[1]You can assume the radial component of **B** is positive,then by reversing the direction of the curren,the radial component of **B** would be negative. However, reversing the current is equivalent to flipping the solenoid, which of course should not change the direction of the radial component. So the radial part should always be zero

where n is the number density of the loops. Since the line segment AB can be chosen arbitrarily inside, we can conclude that the $\mathbf{B}$ field is a constant field inside the solenoid with the magnitude given above.

II. The magnetic field of a long rotating hollow cylinder (HW6 Q12 (b))

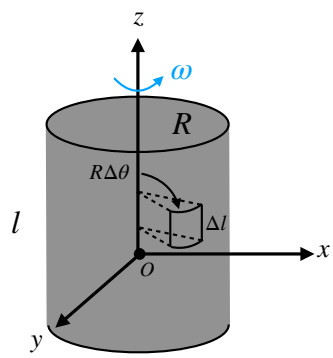

Figure 2: A rotating charged cylinder

We have seen from (a) that the $\mathbf{B}$ field is given by $B = \frac{4\pi n I}{c}$. We can further identify nI as the surface current density K defined via $I = \int \mathbf{K} \cdot d\mathbf{l}$. Therefore,

$$B = \frac{4\pi K}{c}.$$

To calculate the surface current density, we suppose that the cylinder ratates by an angle $\Delta\theta$ during a time element Δt, then the amount of charges that pass through the line Δl is given by

$$\Delta Q = \sigma \Delta A = \sigma(\Delta l)(R\Delta\theta) = \sigma(\Delta l)(R\omega\Delta t)$$

where σ is the charge density on the cylinder. The surface current density thus is

$$K = \frac{\Delta Q}{\Delta t \Delta l} = \sigma R\omega.$$

Then the magnetic field is

$$B = \frac{4\pi K}{c} = \frac{4\pi}{c}\sigma R\omega = \frac{4\pi}{c}\frac{Q}{2\pi R l} \cdot R \cdot 2\pi f = \frac{4\pi}{c}\frac{Qf}{l}.$$

III. The magnetic field in the gap of the ring electromagnet (HW7 Q11)

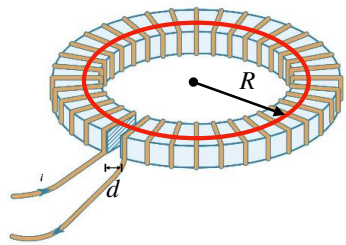

Figure 3: A toroidal coil with a slit

- It is obvious that if the slit is infinitesimally small, the magnitude of the **H** is the same everywhere along the Amperian loop. Then according to the Ampere's circuital law,

$$\oint \mathbf{H} \cdot d\mathbf{l} = \frac{B}{\mu} \cdot 2\pi R = NI$$
$$B = \frac{\mu N I}{2\pi R}$$

  where N is the number of loops enclosed by the Amperian loop.
- If the width of the slit, let's say d, is finite, then we need to divide the loop integral into two pieces:

$$\oint \mathbf{H} \cdot d\mathbf{l} = \frac{B}{\mu} \cdot (2\pi R - d) + \frac{B}{\mu_0} \cdot d = NI$$
$$B = \frac{NI}{\frac{2\pi R - d}{\mu} + \frac{d}{\mu_0}}$$

  with μ and μ_0 being the permeabilities in the media and vacuum, respectively.

Think: In the derivation above, we assumed that the magnitude of the **B** field is the same both in the media and in the slit. Why?